

ScaleOps Core • 30-Day Reliability Blueprint

Distributed Ingestion Queues • PostgreSQL 112x Query Speedup • PgBouncer Defense • Sub-500ms AI Gateway

Client: Canadian SaaS Founder / CTO (Toronto/Vancouver)

Engagement: Staff Augmentation • 20–25 Hrs/Wk Available

Calibrated Rate: \$40.00/hr USD (Client Avg \$50/hr)

Live Prototype: VERIFIED & OPERATIONAL

PRODUCTION SCOPE & 30-DAY MILESTONE DELIVERY SCHEDULE

Live Cockpit: <https://scaleops-core.vercel.app>

MILESTONE	ARCHITECTURE & PRODUCTION ENGINEERING DELIVERABLES	TIMELINE	SHARE	ALLOCATION
Phase 0	Interactive Architecture Prototype & Operational Cockpit (Live) Living demo: 5,000 req/s Redis Stream ingestion buffer, PostgreSQL 112x query acceleration (428ms to 3.8ms), PgBouncer connection multiplexing (58:1 ratio), Redis XFetch stampede defense, and sub-500ms AI circuit breaker.	Live Now	Included	\$0.00 (Live)
Milestone 1	Codebase Discovery, Observability Baseline & Staging CI/CD Audit Audit current backend codebase (FastAPI/Django/Node), profile runtime latency, review Docker/K8s compose configurations, establish Prometheus/Datadog metrics baseline, and ship initial atomic PR to validate CI/CD pipeline.	Week 1	20%	25 Hrs (\$1,000)
Milestone 2	PostgreSQL 16 Slow Query Profiling, Execution Plans & Composite Indexing Analyze EXPLAIN (ANALYZE, BUFFERS) telemetry on top 10 bottleneck queries. Eliminate disk buffer sequential scans via composite covering B-Trees, implement zero-downtime CONCURRENT index migrations, and cut P99 DB latency by 90%+.	Week 2	20%	25 Hrs (\$1,000)
Milestone 3	Asynchronous Webhook Ingestion, Redis 7 Streams & Autoscaled Workers Decouple synchronous HTTP request threads with Redis 7 Streams buffer. Scale Celery/Inngest consumers from 4 to 24 workers under traffic surges, configure exponential backoff retries with full jitter, and wire Dead-Letter Queue (DLQ) alerts.	Week 3	20%	25 Hrs (\$1,000)
Milestone 4	PgBouncer Connection Pooling & Redis XFetch Stampede Elimination Deploy PgBouncer in transaction pooling mode multiplexing 1,450 web connections to 25 PostgreSQL server slots. Implement probabilistic early cache recomputation (XFetch) to permanently eliminate thundering herd cache stampedes on hot keys.	Week 4	20%	25 Hrs (\$1,000)
Milestone 5	Multi-Provider AI Gateway, Rate Limiting & Production Handover Docs Implement sub-500ms automated circuit breaker (Gemini Flash to GPT-4o-mini), inline PII token scrubbing, OWASP LLM01 prompt injection defense, comprehensive integration test suite, and clean architecture runbooks with zero lock-in.	Week 5	20%	25 Hrs (\$1,000)
TOTAL 30-DAY SYSTEMS RELIABILITY ENGAGEMENT (STAFF AUGMENTATION / 20–25 HRS/WK)		30 Days	100%	\$40.00/hr Capped

30-DAY SYSTEMS RELIABILITY ROADMAP

Phase 0: Interactive ScaleOps Cockpit (Delivered) \$0.00 (Live Ahead of Bid)

W1: Codebase Discovery & Observability Baseline 25 Hrs (Net Day 7)

W2: PostgreSQL Profiling & Composite Indexing 25 Hrs (Net Day 14)

W3: Ingestion Decoupling & Worker Autoscaling 25 Hrs (Net Day 21)

W4: PgBouncer Pooling & XFetch Stampede Defense 25 Hrs (Net Day 28)

W5: Multi-Provider AI Gateway & Handover Docs 25 Hrs (Net Day 35)

PRODUCTION ARCHITECTURE & RELIABILITY GUARDRAILS

✓ **Zero Downtime Migrations:** All PostgreSQL indexing deployed CONCURRENTLY with strict lock timeouts.

✓ **Database Connection Shield:** PgBouncer transaction pooling prevents connection pool exhaustion at 5,000 req/s.

✓ **Cache Stampede Defense:** Redis XFetch probabilistic early refresh eliminates database spikes on expired keys.

✓ **Staff Augmentation Hygiene:** Clean atomic PRs, daily async standup notes, 100% test pass rate, zero micromanagement.

✓ **Cloud Cost Right-Sizing:** Optimized query plans & pooling cut AWS RDS and compute bills by up to \$2,450/month.

CALIBRATED HOURLY RATE

\$40.00/hr (20–25 hrs/week). Aligned with client historical average (\$50/hr). Flexible weekly sprint cadence.

CLOUD COST REDUCTION

Halves AWS RDS and serverless compute waste (~\$29,400/yr saved), paying back engineering costs in weeks.

100% CODE OWNERSHIP

All code, PRs, Terraform modules, and Docker configs committed directly to client's Git repositories.

DAILY ASYNC STANDUPS

Daily notes on shipped work, upcoming PRs, blockers, and test status. Zero hand-holding required.

FORMAL AUTHORIZATION & SYSTEMS RELIABILITY ACCEPTANCE

BINDING UPON CONTRACT ACTIVATION VIA UPWORK HOURLY OFFER / MILESTONE SCHEDULE

AUTHORIZED ARCHITECT: BARAKAHSOFT LLC (WYOMING, USA)
Signatory: **Shakil Ahmed** • Principal Systems Architect & Former Lead Engineer at Legiti

AUTHORIZED CLIENT: CANADIAN B2B SAAS PLATFORM
Signatory: **Founder / CTO** • Authorized SaaS Client Representative

Shakil Ahmed

17 Sep 2026

AUTHORIZED ARCHITECT SIGNATURE

DATE

[Accepted via Upwork Contract Offer / Sign-off]

____ / ____ / 2026

AUTHORIZED CLIENT SIGNATURE

DATE